Lecture 14: Cognitive models of semantic representation

PSYC 51.17: Models of language and communication

Jeremy R. Manning
Dartmouth College
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Learning objectives

By the end of this lecture, you will

1. Compare and contrast computational vs. human views of semantic representation
2. Explain the symbol grounding problem and its relevance to large language models
3. Understand the principles of embodied cognition and conceptual metaphors
4. Distinguish between taxonomic similarity and thematic relatedness
5. Evaluate empirical evidence from neuroscience regarding semantic maps in the brain
6. Analyze the "stochastic parrots" debate and the limits of form-based learning

Today's Lecture

Outline

1. **Human vs. Computational Semantics**
2. **Cognitive Theories of Meaning**
3. **Embodied & Grounded Cognition**
4. **Semantic Similarity: What Does It Mean?**
5. **Empirical Evidence from Cognitive Science**
6. **Bridging the Gap: Models Meet Minds**

Goal: Understand what computational models are really learning

The Fundamental Question

What does "meaning" actually mean?

Computational View:

- Vectors in high-dimensional space
- Learned from text co-occurrence
- Distributional patterns
- Statistical relationships
- "You shall know a word by the company it keeps"

Human View

- Sensory experiences
- Emotional associations
- Physical grounding
- Social context
- Multimodal integration
- Embodied understanding

How Do Humans Represent Meaning?

Example: The word "coffee"

What comes to YOUR mind?

- **Visual:** brown liquid, mug, steam, beans
- **Olfactory:** aroma, roasted smell
- **Gustatory:** bitter taste, smooth texture
- **Tactile:** hot, warm cup, liquid
- **Auditory:** brewing sounds, pouring
- **Motor:** lifting cup, drinking motion
- **Contextual:** morning routine, work, café
- **Emotional:** comfort, alertness, pleasure
- **Social:** conversations, meetings
- **Cultural:** Starbucks, espresso, traditions

Word2Vec representation

`coffee = [0.23, -0.45, 0.67, 0.12, -0.89, ... ]` (300 numbers learned from text)

What's missing?

- No sensory grounding, embodied experience, or emotional content
- No perceptual features or action affordances

The Symbol Grounding Problem

Harnad (1990): Can symbols have intrinsic meaning?

The Chinese Room (Searle, 1980):

Input → [Rule Book: If see X output Y] → Output. Correct response! But no understanding of Chinese.

Analogy to LLMs:

Input: "Is a penguin a bird?" → [Pattern: "X is a bird" often follows penguins in text] → Output: "Yes, a penguin is a bird". Correct! But does it "know" birds?

Symbol vs. Grounded Systems

Symbol Systems: Symbols refer to other symbols; purely syntactic manipulation; no connection to world.

Grounded Systems: Symbols connected to perception. A child learns "dog" by SEEING, PETTING, HEARING, and being LICKED by dogs. The word "dog" is grounded in experience!

Further reading

Searle (1980, *Behavioral and Brain Sciences*) Minds, Brains, and Programs.

Harnad (1990, *Physica D*) The Symbol Grounding Problem.

Embodied Cognition Theory

Meaning arises from bodily experience and sensorimotor interaction

Key Principles:

1. **Embodiment:** Cognition shaped by body
2. **Situatedness:** Meaning context-dependent
3. **Enactivism:** Knowing through doing
4. **Grounding:** Concepts tied to perception/action

Neuroscience Evidence

Reading "kick the ball" → Motor cortex activates (Leg area specifically!)
Reading "pick up the cup" → Motor cortex activates (Hand area activates)
Even without moving!

Conceptual Metaphors (Lakoff & Johnson)

Physical → Abstract mapping:

- "WARM personality" ← holding warm drink primes positive judgments!
- "HIGH status" ← up = good, down = bad (heads held high)
- "GRASPING an idea" ← physical grasping simulated mentally
- "Heavy heart" ← weight = emotional burden

These metaphors are NOT in Word2Vec! Models learn word patterns, not embodied experience.

The Distributional Hypothesis Revisited

"You shall know a word by the company it keeps" — J.R. Firth (1957)

Strong version: Word meaning IS distributional patterns
Weak version: Distributional patterns REFLECT meaning

Supports

- Works remarkably well in practice; captures semantic similarity
- Enables analogical reasoning; scales to huge vocabularies
- Unsupervised learning; aligned with usage-based linguistics

Limitations

- Correlation $\neq$ causation; lacks perceptual grounding
- No embodied understanding; missing common sense
- Can't handle novel situations; reflects training data biases

Key Question

Is distributional semantics sufficient for meaning, or just a useful approximation?

Further reading

Boleda (2020, *Annual Review of Linguistics*) Distributional Semantics and Linguistic Theory.

What Does "Semantic Similarity" Really Mean?

Different types of similarity

Taxonomic (IS-A): dog-cat (Both are animals) → Similarity: HIGH
Thematic (GOES-WITH): dog-leash (Co-occur in events) → Relatedness: HIGH, Similarity: LOW!

Test Yourself

Which is more SIMILAR to "coffee"?
A) tea ← Same category (beverages)
B) cup ← Co-occurs (thematic)

Answer: A (tea) is more SIMILAR; B (cup) is more RELATED.

What Word2Vec Says

Word2Vec often gets this wrong!

```
model.similarity('coffee', 'cup') # 0.65
model.similarity('coffee', 'tea') # 0.62
```

Cup ranked higher due to co-occurrence! But tea is categorically more similar.

SimLex-999 vs WordSim-353

Pair	SimLex	WordSim
car-auto	0.96	0.92

Semantic Projection: Recovering Human Knowledge

Grand et al. (2022): Can we extract human-like features from embeddings?

The Experiment:

1. Collect human ratings on perceptual features (edible, heavy, alive, holdable, etc.)
2. Train linear projections from word embeddings
3. Test if embeddings predict human ratings

Key Findings

- Embeddings encode surprisingly rich knowledge
- Can predict perceptual features (even vision and motor properties!)
- Better with larger models

Interpretation

Text co-occurrence captures real-world properties through indirect grounding in language. But still not true perceptual grounding.

Further reading

Grand et al. (2022, *Nature Human Behaviour*) Semantic projection recovers rich human knowledge of multiple object features from word embeddings.

Multimodal Models: Bridging the Gap

Combining language with perception

Vision-Language Models: CLIP (OpenAI), ALIGN (Google), Flamingo (DeepMind), GPT-4V (OpenAI)

Key Idea: Learn joint embedding space where text and images map to same space. Enables cross-modal understanding and grounds language in vision.

Training

- Image-caption pairs; Contrastive learning
- Matching images to descriptions
- Large-scale (400M+ pairs)

Benefits

Visual grounding, zero-shot classification, better generalization, more "human-like".

Further reading

Radford et al. (2021, *ICML*) Learning Transferable Visual Models From Natural Language Supervision (CLIP).

Conceptual Spaces Theory

Gärdenfors (2000): Meaning as geometry

Key Ideas:

- Concepts represented in quality dimensions (color, size, temperature, etc.)
- Each dimension has a metric; Concepts are regions in space
- Similarity = geometric proximity

Example: Colors

- Hue, saturation, brightness
- Natural categories (red, blue, green) with fuzzy boundaries
- Prototypes at centers

Relation to Embeddings

- Similar geometric structure but learned dimensions
- Not interpretable qualities; No grounding in perception

Further reading

Gärdenfors (2000, *MIT Press*) Conceptual Spaces: The Geometry of Thought.

Lexical Semantic Theories

How do linguists think about word meaning?

1. **Feature-Based:** Words = bundles of features (e.g., bachelor = [+human, +male, +adult, -married])
2. **Prototype Theory:** Categories have best examples (Robin is prototypical; Penguin is peripheral)
3. **Frame Semantics:** Words evoke conceptual frames (e.g., "Buy" activates commerce frame)
4. **Construction Grammar:** Meaning from form-function pairings; usage-based
5. **Word Sense Disambiguation:** Words have multiple senses (WordNet: Bank_1 vs Bank_2)

Question

Which theories align with distributional models?
Answer: Mostly usage-based views (Construction Grammar, Prototype Theory). Models struggle with feature analysis and frame semantics.

Further reading

Fillmore (1982, *The Linguistic Society of Korea*) Frame Semantics.

Rosch (1975, *Journal of Experimental Psychology: General*) Cognitive Representations of Semantic Categories.

Empirical Evidence from Neuroscience

What does the brain tell us about semantic representation?

fMRI Studies: Can predict brain activity from word embeddings. Semantic information is distributed across cortex in different regions for different features (Temporal: objects; Motor: actions; Visual: visual features).

Findings

- Word2Vec correlates with neural patterns, but imperfectly
- Human brain uses multimodal integration (Language areas + sensory areas)

Brain vs. Model Representations

Feature	Brain	Model (BERT)
Distributed	Yes	Yes
Hierarchical	Yes	Yes
Context-sensitive	Yes	Yes
Grounded	Yes	No
Fast	Yes	Yes

Further reading

The "Stochastic Parrots" Debate

Bender et al. (2021): On the Dangers of Stochastic Parrots

The Argument: LLMs learn form, not meaning. They are "stochastic parrots" repeating patterns without understanding of the world or communicative intent. Risk: Mistaking fluency for understanding.

Evidence: Fail on simple reasoning; sensitive to phrasing; hallucinate facts; no common sense; brittle to adversarial inputs.

Counter-Arguments

Emergent capabilities at scale; performance on complex tasks; transfer to new domains. Maybe understanding = prediction? Pragmatic success criterion.

The Core Question

Can meaning arise from form alone? Or do we need grounding in perception, action, social interaction, and physical embodiment?

Further reading

Bender & Koller (2020, *ACL*) Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data.

Bender et al. (2021, *FAccT*) On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

Common Sense Reasoning

What humans know but models don't

Physical Intuition Failures:
Q: "Can you fit an elephant in a refrigerator?"
GPT-3: "Yes, if you open the door wide enough..."

Winograd Schema (reasoning):
"The trophy doesn't fit in the brown suitcase because it is too [small/large]."
What does "it" refer to? "small" → suitcase; "large" → trophy. Requires world knowledge!

Social Intuition Failures

Q: "John told Mary he loved her. How did Mary feel?"
Depends on context: First date? (Surprised/happy); After argument? (Relieved); Unwanted? (Uncomfortable). Models miss social nuance!

Why Models Struggle

Physical/social knowledge is rarely stated; assumed as background knowledge. Requires embodied experience and causal reasoning.

Compositionality: Phrases and Sentences

How do we combine word meanings?

The Problem: Vector math doesn't work!

`vec("hot") + vec("dog") ≠ vec("hot dog")`.

"hot dog" = food item; "hot" + "dog" = warm canine.

Same issue: `vec("red") + vec("herring") ≠ vec("red herring")` (distraction, not a fish!).

Non-compositional Phrases: Kick the bucket (= die); Spill the beans (= reveal secret); Break a leg (= good luck).

What Transformers Learn

Input: "break a leg" → Attention sees this phrase often in "good luck" contexts → Output: idiomatic meaning. But fails on novel combinations!

The Negation Problem: BERT struggles with negation. "The movie was good" vs "The movie was not good" embeddings are very similar (~0.85 cosine sim), but "not" should flip the meaning.

Grand Discussion

Do large language models 'understand' language?

Arguments FOR:

- Solve complex tasks; Generalize to new domains; Emergent capabilities
- Capture linguistic structure; Pragmatic criterion: if it works...
- Maybe understanding = prediction; Human understanding also imperfect

"The question is not whether machines think, but whether they behave intelligently" — Turing

Arguments AGAINST

- No grounding in reality; No embodied experience; No intentionality
- Brittle, exploit shortcuts; Hallucinate confidently; No causal reasoning
- Missing common sense; Form without meaning

"Understanding requires grounding in perception and action" — Embodied cognition

Perhaps the wrong question?

Instead of "Do they understand?", ask: What do they represent? How does it differ from humans? What are the limits?

Summary

What we learned today

1. **Symbol Grounding:** Computational models lack perceptual grounding
2. **Embodied Cognition:** Human meaning tied to bodily experience
3. **Distributional Semantics:** Powerful but incomplete theory
4. **Semantic Similarity:** Multiple types, models capture some
5. **Empirical Evidence:** Models align with neural patterns but miss multimodality
6. **Common Sense:** Models struggle with physical and social reasoning
7. **Compositionality:** Non-literal language remains challenging

The gap between computation and cognition remains, but we're making progress!

Questions?

Email me | Join our Discord | Come to office hours

Coming up: Attention mechanisms and the transformer revolution!